

SPAN 580 - Applied Spanish Phonetics
Spring 2011 - University of Arizona, Tucson

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Meeting hours: M 3:00pm - 5:30pm

Office hours: MW 10:00am-10:50am, W 10:00am-10:50am; 3:00pm-5:00pm

Course description

This course serves as an introduction to the study of speech from the perspective of linguistic phonetics. In particular, it serves as an introduction to the study of Spanish phonetics. The study of the sounds of Spanish will be approached from the perspective of experimental, instrumental, quantitative phonetics. As such, we will investigate the production (articulatory phonetics), transmission (acoustic phonetics) and audition and processing (perceptual phonetics) of Spanish sounds. Additionally, this course assumes an “applied” perspective in that it will help students to explore issues pertaining to bilingualism, second language learning, dialectal variation and sound change.

An important objective of this course is to help students to acquire the skills necessary to carry out experimental speech research, including issues such as hypothesis testing, experimental design, quantitative data collection and archiving, quantitative analysis and, finally, interpretation of the implications of experimental results. Students will learn to use fundamental tools such as Praat (digital speech processing and behavioral experimental design) and R (statistics, mathematics and graphics). Both software programs are free and available for both Mac OS X, Windows and Linux. Students are required to download and install these programs onto their personal computers. Students will be asked to bring their personal computers to some class meetings. This will be done in advance, over electronic mail.

Materials

RJ. Phonetics: Transcription, production, acoustics and perception. H. Reetz & A. Jongman. Wiley-Blackwell. 2008. Malden, MA., Oxford, UK., West Sussex, UK.

Package of additional readings to be distributed in electronic format (.pdf) as they become necessary. Most readings are from peer-reviewed journal articles.

Main journals: *Journal of Phonetics*, *Phonetica*, *Journal of the Acoustical Society of America*, *Journal of the International Phonetic Association*, *Language & Speech*, *Speech Communication*, *Cognition*, among others.

Coursework

Students are responsible for reading the materials before the corresponding class meetings. They are expected to participate actively in class discussions by asking questions or adding comments, including criticisms and personal ideas. Students will be responsible for presenting several journal articles in front of the class and leading the discussion of those articles. They will participate in class discussions even when another student is responsible for presenting and leading the discussion of an article.

Required readings other than the textbook (e.g. journal articles) will be evenly distributed among the students on a first-come-first-serve basis. Every student will prepare a standardized handout of the article/s s/he chooses to present. S/he will bring enough copies of the handout for the class. S/he will summarize the topic of the article/s, its methods, its results and its conclusions. Afterwards, s/he will lead the discussion on the article s/he presented. The other students are required to participate in such discussions by asking questions, offering comments, making suggestions to improve the quality of the article, explaining how the article could be replicated by them, etc. All students are required to read all journal articles (and textbook chapters), even those they will not present. Optional readings will be specifically marked as such. Students should assume all readings to be mandatory unless evidence to the contrary is obvious.

Handouts need to be written in Spanish.

Course projects

Every student will be required to turn in one experimental report by the end of the semester: (1) one on speech production (acoustic analysis) or (2) one on speech perception (perceptual identification, discrimination, category-goodness, etc.). These will be four-page reports following the format of the Proceedings of ICPHS (see <http://www.icphs2007.de/> >> Proceedings >> Contents). Students may work in pairs if they so desire. If they work in pairs, they need to submit two papers. That is, students may choose to submit two co-authored reports or one single-authored report. Students can get the Word Template for ICPHS Proceedings at:

<https://sites.google.com/site/miquelsimonet/sports-lab-az/resources> >> Attachments

The studies might be partial replications of published work. The goal of these projects is for the students to learn to use the necessary tools to carry out a research project from its inception to its conclusion. The goal is not to come up with original research ideas. Pilot data (e.g., small group of participants) are generally acceptable. The students are responsible for learning to use the necessary tools (speech recorders, Praat, Praat

scripts, R commands) and are strongly encouraged to explore the tools on their own or with the help of their peers from the beginning of the semester. The students need to explore the use of software on their own (perhaps by searching online). Up to two class sessions may be dedicated to Praat and R after introduction of the main theoretical concepts. Although students might consult with the instructor, they will not expect the instructor to write Praat or R code for them or carry out data analysis with/for them. The instructor might be contacted during office hours for such consultations. Electronic mails are acceptable for all other sorts of consultations, but lengthy private explanations on how to use the software will not be provided through electronic mail.

Final projects need to be written in English. Students will give an oral presentation of their projects during the last class meeting of the semester.

Evaluation

Project	33.3%
Presentations	33.3%
Participation	33.3%

Attendance policy

Regular class attendance is essential for successful completion of the course. More than 1 absences will have a negative effect on your final grade. The 2nd absence and every subsequent absence after that will result in the loss of 5% point off the final overall course grade, regardless of reason. Keep in mind that while you have 1 “free” absence, on the day/s you miss you will not be able to earn participation points, you will miss the material given in class and you might miss your own presentation. If you are absent, contact a classmate immediately to get the assignments and to keep up with the material scheduled in the syllabus. The instructor is not responsible to put you up to date. Do not send emails to the instructor asking for updates if you missed class.

Any planned absence that you are aware of ahead of time, such as religious holidays recognized by the University of Arizona or Dean’s excuses, should be made up before the absence occurs. If you know that you will be absent, it is your responsibility to let the instructor know ahead of time. All holidays or special events observed by any religion will be honored for those students who show affiliation with that particular religion. Absences pre-approved by the UA Dean of Students (or Dean's designee) will be honored.

Code of academic integrity

The instructor will initiate an academic integrity case against students suspected of cheating, plagiarizing, or aiding others in dishonest academic behavior. Students are

responsible for reading and understanding the Code of Academic Integrity.

<http://w3.arizona.edu/~studpubs/handbook/policy.html>

Examples of academic dishonesty include, but are not limited to, plagiarism, cheating, and aiding and abetting dishonesty. An example of plagiarism would be to submit a written sample which in part or in whole is not the student's own work without attributing the source. Cheating includes allowing another person to do your work and to submit the work under one's own name. Any work which is submitted for a grade must be 100% the student's own work. If you are not sure when it is appropriate to seek help, please see your instructor.

<http://dos.web.arizona.edu/uapolicies>

Extra credit

No possibility of individual extra credit exists.

Policies against threatening behavior by students

Any behavior that is seen as disruptive to the classroom environment or is threatening in any way will not be tolerated. Please see the Code of Academic Integrity and the Code of Student Conduct online at:

<http://dos.web.arizona.edu/uapolicies/>

Students with disabilities

Students with disabilities must be registered with the Disability Resource Center, and must submit appropriate documentation to the instructor when requesting reasonable accommodations for class activities.

<http://drc.arizona.edu/instructor/syllabusstatement>

Tentative calendar

s00. [Jan. 17] Holiday. Please, read syllabus and start reading the materials for next week.

s01. [Jan. 24] RJ Chapters 1 (Phonetics in a Nutshell, etc.), 2 (Articulatory Phonetics), 3 (Phonetic Transcription).

s02. [Jan. 31] RJ Chapters 4 (Place and Manner of Articulation of Consonants and Vowels), 5 (Physiology of the Vocal Apparatus), 6 (Airstream Mechanisms and Phonation Types).

a) Lisker, L., & Abramson, A.S (1964). A cross-language study of voicing in initial stops: Acoustical measurements. *Word* 20, 384-422.

s03. [Feb. 7] Co-articulation. Methods in articulatory phonetics.

- a) Recasens, D. & Espinosa, A. (2005) Articulatory, positional and coarticulatory characteristics for clear /l/ and dark /l/: Evidence from two Catalan dialects, *Journal of the International Phonetic Association*, 35, 1-25.
- b) Recasens, D. & Espinosa, A. (2009) An articulatory investigation of lingual coarticulatory resistance and aggressiveness for consonants and vowels in Catalan, *Journal of the Acoustical Society of America*, 125, 2288-2298
- c) Recasens, D. & Espinosa, A. (2010) Lingual kinematics and coarticulation for alveolopalatal and velar consonants in Catalan, *Journal of the Acoustical Society of America*, 127, 3154-3165.

s04. [Feb. 14] RJ Chapters 7 (Basic Acoustics), 8 (Analysis Methods for Speech Sounds). Co-articulation, continued.

s05. [Feb. 21] RJ Chapters 9 (The Source-Filter Theory of Speech Production), 10 (Acoustic Characteristics of Speech Sounds). Vowels, I. Native and non-native Spanish vowels.

- a) Bradlow, A. R. (1995). A comparative acoustic study of English and Spanish vowels. *Journal of the Acoustical Society of America* 97, 1916-1924
- b) Menke, M. & Face, T. (2010) Second language Spanish vowel production: An acoustic analysis. *Studies in Hispanic and Lusophone Linguistics* 3, 181-214.

s06. [Feb. 28] Vowels, II. Theories and methods. The sociophonetics of vowel production.

- a) Recasens, D. & Espinosa, A. Dispersion and variability of Catalan vowels, *Speech Communication* 48, 645-666.
- b) Escudero, P., Boersma, P., Rauber, A. & Bion, R. (2009): A cross-dialect acoustic description of vowels: Brazilian and European Portuguese. *Journal of the Acoustical Society of America* 126, 1379-1393.
- c) Maclagan, M. & Hay, J. (2007) Getting fed up with our feet: Contrast Maintenance and the New Zealand English “short” front vowel shift. *Language Variation and Change*, 19, 1-25

s07. [Mar. 7] Introduction to second language speech learning. Acquisition of vowels.

- a) Flege, J. (2007). Language contact in bilingualism: Phonetic system interactions. *Laboratory Phonology* 9, ed. by Cole, J. & Hualde, J.I. Berlin: Mouton de Gruyter.
- b) Guion, S.G. (2003). The vowel systems of Quichua-Spanish bilinguals: An investigation into age of acquisition effects on the mutual influence of the first and second languages. *Phonetica* 60, 98-128.
- c) Piske, T., Flege, J., MacKay, I., & Meador, D. (2002). The production of English vowels by fluent early and late Italian-English bilinguals. *Phonetica*, 59, 49-71.

s08. [Mar. 21] Stop consonants. Voicing contrasts. Second language learning of stops.

- a) Kang, K.-H. & Guion, S.G. (2006). Phonological systems in bilinguals: Age of learning effects on the stop consonant systems of Korean-English bilinguals. *Journal of the Acoustical Society of America* 119, 1672-1683.
- b) Zampini, M. & Green, K. (2001) The voicing contrast in English and Spanish: The relationship

between perception and production. *One Mind, Two Languages*, ed. by Nicol, J., Oxford: Blackwell.

c) Magloire, J. & Green, K. (1999) A cross-language comparison of speaking rate effects on the production of voice onset time in English and Spanish. *Phonetica* 56, 158-185.

s09. [Mar. 28] Stop consonants. VOTs and Spirantization.

a) Fowler, C., Sramko, V., Ostry, D., Rowland, S., Hallé, P. (2008) Cross language phonetic influences on the speech of French–English bilinguals. *Journal of Phonetics* 36, 649-663.

b) Face, T. & M. Menke. (2009) Acquisition of the Spanish voiced spirants by second language learners. In Collentine, J. (ed.) *Selected Proceedings of the 11th Hispanic Linguistics Symposium*. Somerville, MA: Cascadilla. 39-52.

c) Zampini, M. (1994) The role of native language transfer and task formality in the acquisition of Spanish spirantization. *Hispania* 77, 470-481.

s10. [Apr. 4] Fricatives: Acoustics, articulation, perception.

a) Jongman, A., Wayland, R. & S. Wong (2000) Acoustic characteristics of English fricatives. *Journal of the Acoustical Society of America* 108: 1252-1263.

b) Recasens, D. & A. Espinosa (2007) An electropalatographic and acoustic study of affricates and fricatives in two Catalan dialects. *Journal of the International Phonetic Association* 37: 143-172.

c) Jongman, A., Wang, Y., and Kim, B. (2003). Contributions of sentential and facial information to perception of fricatives. *Journal of Speech, Language, and Hearing Research* 46, 1367-1377.

s11. [Apr. 11] RJ Chapter 11 (Syllables and Suprasegmentals). Prosody:

a) Prieto, P. & F. Torreira (2007) The segmental anchoring hypothesis revisited: Syllable structure and speech rate effects on peak timing in Spanish. *Journal of Phonetics* 35: 473-500.

b) Ortega-Llebaria, M. and Prieto, P. (2011) Acoustic correlates of stress in Central Catalan and Castilian Spanish. *Language and Speech* 54, (1)

c) Mennen, I. (2004). Bi-directional interference in the intonation of Dutch speakers of Greek. *Journal of Phonetics* 32, 543-563.

s12. [Apr. 81] RJ Chapters 12 (Physiology and Psychophysics of Hearing), 13 (Speech Perception). Perception in bilinguals. The role of perception in sound change.

a) Guion, S.G., Flege, J.E., Akahane-Yamada, R. & Pruitt, J. (2000). An investigation of current models of second language speech perception: The case of Japanese adults' perception of English consonants. *Journal of the Acoustical Society of America* 107, 2711-272.

b) Guion, S.G. (1998) The role of perception in the sound change of velar palatalization. *Phonetica* 55, 18-52.

c) Hay, J., Warren, P. & Drager, K. (2006) Factors influencing speech perception in the context of a merger-in-progress. *Journal of Phonetics*, 34, 458-484

s13 [Apr. 25] Speech perception, II:

a) Hojen, A. and Flege, J. (2006). Early learners' discrimination of second-language (L2) vowels. *Journal of the Acoustical Society of America* 119,3072-3084.

b) Flege, J. & MacKay, I. (2004). Perceiving vowels in a second language. *Studies in Second Language Acquisition* 26, 1-34.

c) Aoyama, K., Flege, J., Guion, S., Yamada, T., & Akahane-Yamada, R. (2004). Perceived phonetic dissimilarity and L2 speech learning: The case of Japanese /r/ and English /r/ and /l/. *Journal of Phonetics* 23, 233-250.

s14 [May 2] Speech perception, III:

a) Pallier, C., Bosch, L. & Sebastián-Gallés, N. (1997) A limit on behavioral plasticity in speech perception. *Cognition* 64: B9-B17.

b) Sebastián-Gallés, N. & Soto-Faraco, S. (1999) On-line processing of native and non-native phonemic contrasts. *Cognition* 72, 11-123.

c) Skoruppa, K., Pons, F., Christophe, A., Bosch, L., Dupoux, E., Sebastián-Gallés, N., Alves Limissuri, R., Peperkamp, S. (2009). Language-specific stress perception by nine-month-old French and Spanish infants. *Developmental Science*, 12, 914-919

d) Ramon-Casas, M., Swingle, D., Sebastián-Gallés, N. & Bosch, L. (2009). Vowel categorization during word recognition in bilingual toddlers. *Cognitive Psychology*, 59, 96-121

e) Sebastián-Gallés, N., Echeverría, S. & Bosch, L. (2005) The influence of initial exposure on lexical representation: Comparing early and simultaneous bilinguals. *Journal of Memory and Language* 52, 240-255.